

FJFI - Department of Physics
02MECH - Mechanics
Test 2

13.02.2024
10:00

Name					
Problem	1	2	3	4	Total
Score					

Instructions

1. The time allocated for the exam is 120 minutes.
2. During the exam, you are not allowed to use any class materials or ask your neighbor for help.
3. Calculators are not allowed. You may only have writing supplies and a transparent water bottle without labels on your desk.
4. Use only pens or non-erasable ink pens for the test; erasable pens are not allowed.
5. Write your answers legibly. Illegible answers will not be scored.
6. The maximum score for the test is 4 points.
7. Each question is worth 1 point, but the questions may not require the same amount of time. Therefore, it is advisable to read the entire assignment before starting to solve it.
8. When grading the test, only the solutions provided on the distributed sheets will be considered.
9. Always provide your solutions in separate sheets.

Question 1

Consider a system of three point particles, each having an identical mass m , with their positions defined by time-dependent vectors,

$$\vec{r}_1 = (at^2, vt, 0), \vec{r}_2 = (vt, at^2, b), \vec{r}_3 = (0, b, at^2),$$

where a , b , and v are positive constants. Determine the total external force and the total torque exerted on this system concerning the origin of coordinates. Additionally, based on these findings, analyze whether the system conserves linear and angular momentum.

Solution:

Question 2

Two boats, each having a total mass M , are moving in the same direction at a uniform velocity v . At a certain point, a package of mass m is thrown from the front boat to the rear boat, with a velocity u relative to the front boat. Determine the resulting velocities of both boats and their relative velocity after the package is thrown. Additionally, analyze whether this inelastic process might lead to a collision between the boats.

Solution:

Question 3

Determine the moment of inertia for a thin non-homogeneous rod of length l . The rod's linear mass density exhibits a linear increase from λ_1 to λ_2 along its length. Calculate the moment of inertia concerning an axis passing through the rod's center and perpendicular to its length.

Solution:

Question 4

Calculate the acceleration of a uniform loop as it rolls without sliding on a rough inclined plane inclined at an angle α to the horizontal, in a uniform gravitational field of magnitude g .

Solution: